

Hospital Performance Analytics Dashboard using Power BI

Project Title

Hospital Performance Analytics Dashboard using Microsoft Power BI

Prepared By

Madhavan, Omkar Shendure, Gaurav Sharma, Krish Patel, Govind, Manish, Nitin

Tools Used

- Microsoft Power BI
- Power Query
- DAX
- CSV Dataset

Project Duration

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Introduction

Healthcare organizations generate large amounts of performance data across different hospitals and quality measures. Decision-makers require interactive dashboards to monitor hospital performance, identify low-performing hospitals, compare states, and simulate future improvement scenarios.

This project develops an interactive Power BI dashboard using publicly available hospital performance data. The dashboard enables users to explore hospital quality, geographic distribution, and performance indicators while supporting scenario-based analysis through What-If Parameters.

Business Problem

Hospital administrators often struggle to answer questions such as:

1. Which hospitals perform best?
2. Which states have the highest-rated hospitals?
3. Which quality domain requires improvement?
4. How many hospitals satisfy a minimum rating threshold?
5. How would improving Safety or Readmission affect performance?

Traditional spreadsheet reports are static and difficult to analyze. The objective is to replace static reports with an interactive business intelligence solution.

Project Objectives

The dashboard was developed to:

6. Analyze overall hospital performance.
7. Compare hospitals across states.
8. Evaluate Mortality, Safety and Readmission measures.
9. Identify high-performing hospitals.
10. Perform scenario analysis using What-If Parameters.
11. Improve decision making through interactive reporting.

Dataset Description

Important Columns

Column Name	Description
FacilityID	Unique identifier for each hospital
FacilityName	Name of the hospital facility
Address	Street address of the hospital
City	City where the hospital is located
State	Two-letter state abbreviation
County	County or parish name
HospitalType	Type of hospital (e.g., Acute Care, Critical Access)
HospitalOwnership	Ownership category (e.g., Non-Profit, Government, For-Profit)
EmergencyServices	Whether emergency services are available (Yes/No)
OverallRating	Hospital quality rating on a scale of 1 to 5
BirthingFriendly	Whether the hospital meets birthing-friendly designation
Mortality (Mort)	Mortality domain performance measures (Better/Worse/NoDifferent)
Safety	Safety domain performance measures (Better/Worse/NoDifferent)
Readmission (Readm)	Readmission domain performance measures (Better/Worse/NoDifferent)

Dataset Summary

Attribute	Detail
Source	Hospital General Information (CSV)
Total Records	5,384 Hospitals
Number of Columns	38
File Format	CSV
Primary Key	Facility ID
Geographic Coverage	United States
Rating Scale	1 to 5

Data Cleaning & Transformation

Power Query Transformations

The following transformations were performed:

12. Imported CSV dataset
13. Renamed columns for consistency

14. Corrected data types
15. Removed duplicates
16. Replaced missing values
17. Created lookup tables (State, HospitalType, HospitalOwnership)
18. Built relationships between tables
19. Loaded cleaned data into Power BI

Benefits

- Improved model performance
- Better filtering capabilities
- Reduced redundancy
- Faster dashboard response

Data Model (Entity Relationship Diagram)

The data model was designed using a **star schema** to improve query performance and simplify reporting. The Hospital table serves as the central fact table, connected to lookup tables such as State, Hospital Type, and Hospital Ownership. Relationships are configured as many-to-one with single-direction filtering to ensure accurate aggregations and efficient report performance.

Tables and Relationships

From (One Side)	To (Many Side)	Key
State (1)	Hospital (*)	StateID
HospitalType (1)	Hospital (*)	HospitalTypeID
HospitalOwnership (1)	Hospital (*)	HospitalOwnershipID
Hospital (1)	PerformanceDomain (*)	FacilityID

Table: State

Column	Description
StateID	Primary Key
State	Two-letter abbreviation
State Full Name	Full state name

Table: Hospital (Central)

Column	Description
FacilityID	Primary Key
FacilityName	Hospital name
Address	Street address
City	City name
County	County/Parish
BirthingFriendly	Yes/No
EmergencyServices	Yes/No

Table: HospitalType

Column	Description
HospitalTypeID	Primary Key
Hospital Type	Display label
HospitalType	Internal code

Table: HospitalOwnership

Column	Description
HospitalOwnershipID	Primary Key
HospitalOwnership	Internal code
Ownership	Display label

Table: PerformanceDomain

Column	Description
FacilityID	Foreign Key to Hospital
Domain	Mort, Safety, or Readm
Metric	Better, Worse, NoDifferent, FacilityMeasureCount
Value	Numeric score (SUM aggregation)

Page 1: Executive Overview

DAX Measures Used

1. Total Hospitals

Purpose: Counts the total number of hospitals.

```
Total Hospitals = DISTINCTCOUNT(Hospital[FacilityID])
```

2. Hospitals Above Threshold

Purpose: Counts hospitals meeting the selected rating threshold.

```
Hospitals Above Threshold = CALCULATE(DISTINCTCOUNT(Hospital[FacilityID]),  
FILTER(Hospital, Hospital[OverallRating] >= [RatingThreshold Value]))
```

3. % High Performing

Purpose: Calculates the percentage of hospitals above the threshold.

```
% High Performing = DIVIDE([Hospitals Above Threshold], [Total Hospitals])
```

4. Average Better %

```
Average Better % = DIVIDE([Safety Better %] + [Readm Better %], 2)
```

5. Avg Overall Rating

```
Avg Overall Rating = AVERAGE(Hospital[OverallRating])
```

6. Birthing Friendly %

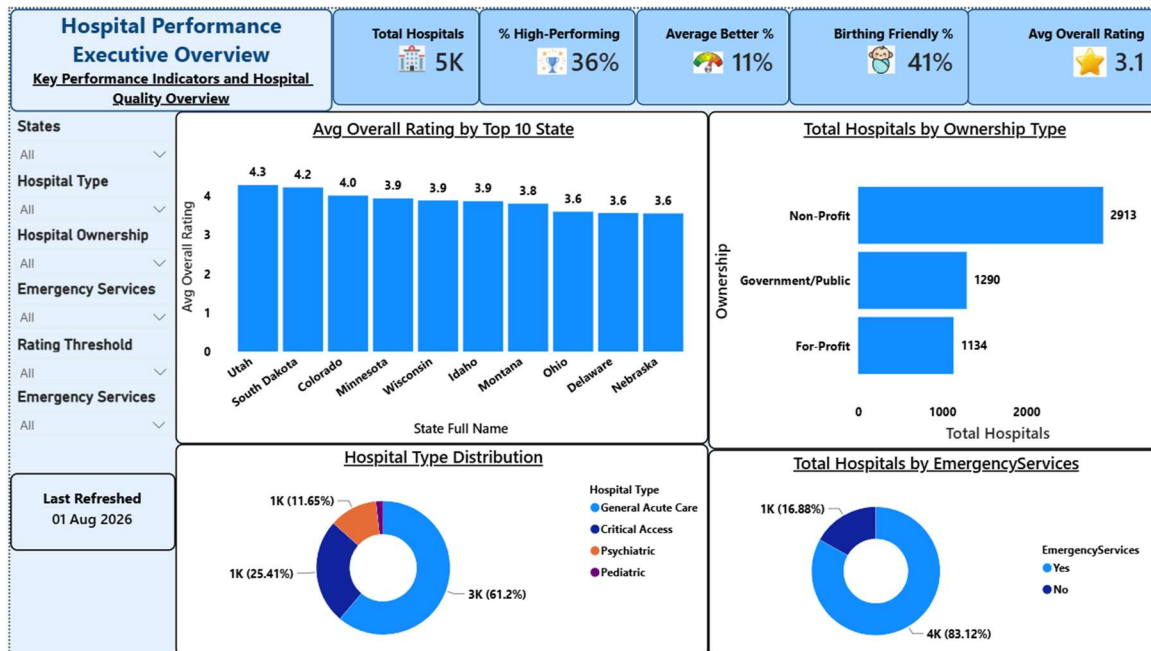
```
Birthing Friendly % = DIVIDE(CALCULATE(COUNTROWS(Hospital),  
Hospital[BirthingFriendly]="Yes"), [Total Hospitals])
```

Features

- Total Hospitals KPI (~5K)
- High Performing Hospitals (36%)
- Average Better % (11%)
- Birthing Friendly % (41%)

- Average Overall Rating (3.1)
- Emergency Services Distribution (83% Yes, 17% No)
- Ownership Distribution (Non-Profit: 2,913 | Government: 1,290 | For-Profit: 1,134)
- Hospital Type Distribution (Acute Care: 61.2% | Critical Access: 25.41% | Psychiatric: 11.65%)
- State-wise Ratings (Top: Utah 4.3, South Dakota 4.2, Colorado 4.0)

Business Insight: This page provides executives with a quick snapshot of hospital quality across the United States.

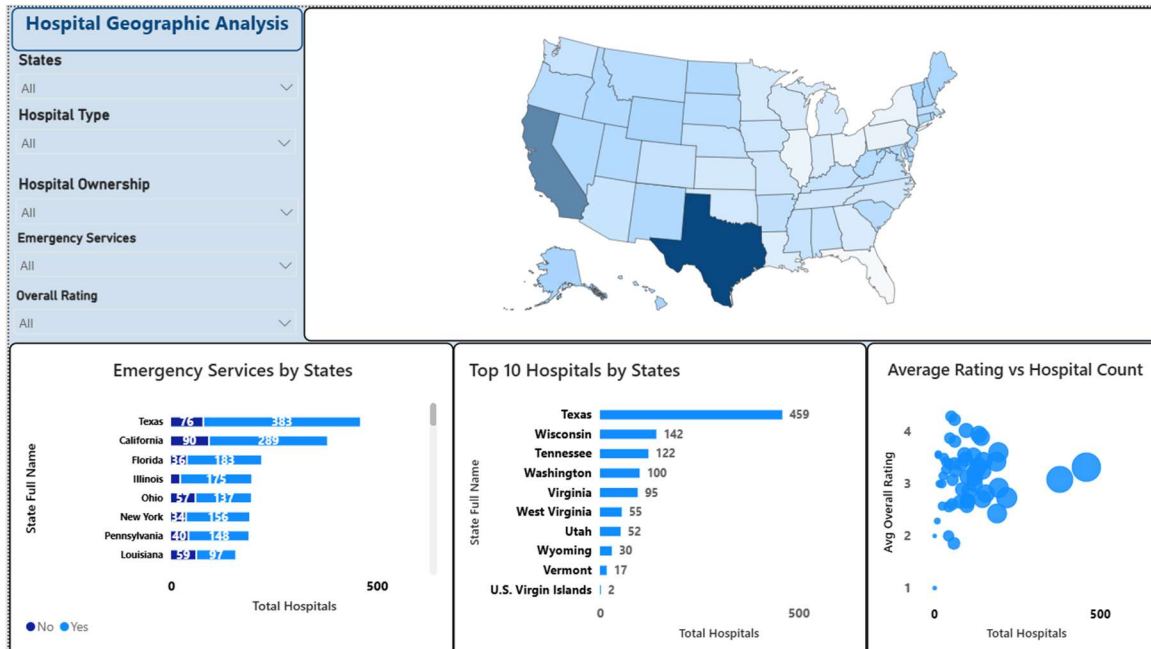


Page 2: Geographic Analysis

Features

- Interactive Map showing hospital distribution across states
- Custom Tooltips with hospital-level detail on hover
- Top 10 Hospitals by State (Texas: 459 | Wisconsin: 142 | Tennessee: 122)
- Emergency Services by State
- Average Rating vs Hospital Count Scatter Chart

Business Insight: Users can quickly identify regional performance differences and geographic concentration of hospitals.



Page 3: Performance Measure Details

DAX Measures Used

1. Overall Better %

Overall Better % = $\text{DIVIDE}([\text{Mort Better \%}] + [\text{Safety Better \%}] + [\text{Readm Better \%}], 3)$

2. Mort Better %

Mort Better % = $\text{DIVIDE}(\text{CALCULATE}(\text{SUM}(\text{PerformanceDomain}[\text{Value}]), \text{PerformanceDomain}[\text{Domain}] = \text{"Mort"}, \text{PerformanceDomain}[\text{Metric}] = \text{"Better"}), \text{CALCULATE}(\text{SUM}(\text{PerformanceDomain}[\text{Value}]), \text{PerformanceDomain}[\text{Domain}] = \text{"Mort"}, \text{PerformanceDomain}[\text{Metric}] = \text{"FacilityMeasureCount"}), 0)$

3. Safety Better %

Safety Better % = $\text{DIVIDE}(\text{CALCULATE}(\text{SUM}(\text{PerformanceDomain}[\text{Value}]), \text{PerformanceDomain}[\text{Domain}] = \text{"Safety"}, \text{PerformanceDomain}[\text{Metric}] = \text{"Better"}), \text{CALCULATE}(\text{SUM}(\text{PerformanceDomain}[\text{Value}]), \text{PerformanceDomain}[\text{Domain}] = \text{"Safety"}, \text{PerformanceDomain}[\text{Metric}] = \text{"FacilityMeasureCount"}), 0)$

4. Readm Better %

Readm Better % = $\text{DIVIDE}(\text{CALCULATE}(\text{SUM}(\text{PerformanceDomain}[\text{Value}]), \text{PerformanceDomain}[\text{Domain}] = \text{"Readm"}, \text{PerformanceDomain}[\text{Metric}] = \text{"Better"}), \text{CALCULATE}(\text{SUM}(\text{PerformanceDomain}[\text{Value}]), \text{PerformanceDomain}[\text{Domain}] = \text{"Readm"}, \text{PerformanceDomain}[\text{Metric}] = \text{"FacilityMeasureCount"}), 0)$

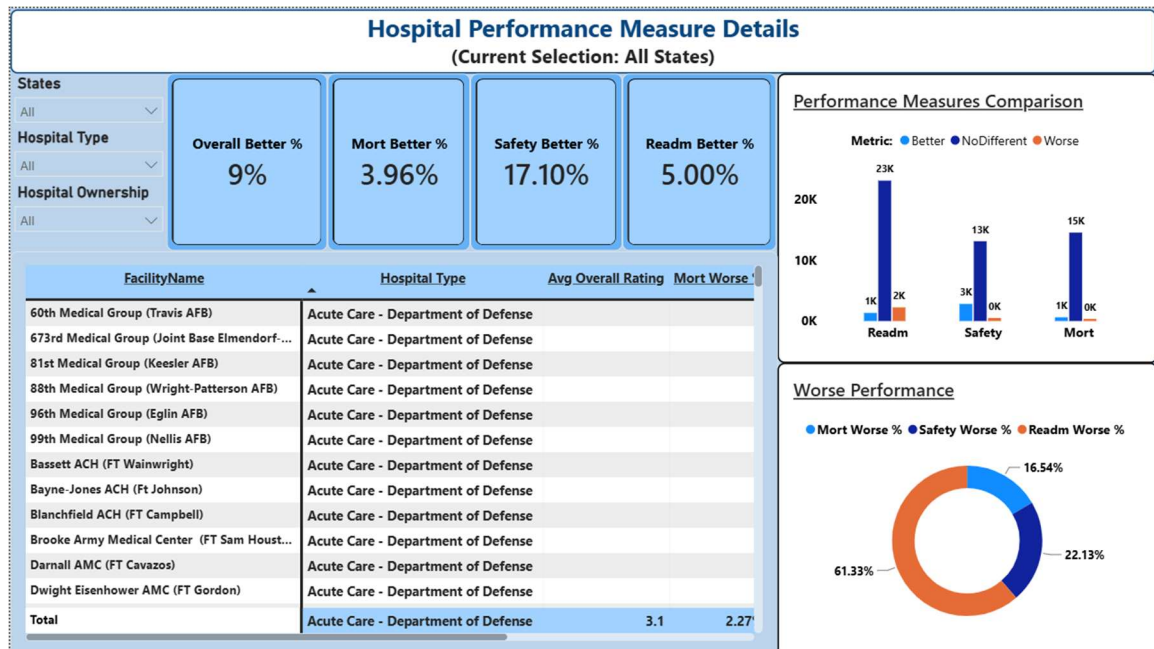
Key Metrics

Domain	Better %	Worse %
Safety	19.61%	22.13%
Readmission	5.00%	16.54%
Mortality	3.96%	61.33%
Overall	9%	—

Features

- Performance KPI Cards (Overall Better 9%, Mort 3.96%, Safety 17.10%, Readm 5.00%)
- Matrix Visual with hospital-level detail
- Performance Measures Comparison (Clustered Bar Chart: Better vs NoDifferent vs Worse)
- Worse Performance Analysis (Donut Chart: Mortality 61.33%, Safety 22.13%, Readm 16.54%)
- Drill-through to Hospital Details

Business Insight: Hospital performance can be analyzed at an individual hospital level using multiple quality measures. Mortality is the weakest domain with 61.33% of measures in the "Worse" category.



Page 4: Scenario Insights

DAX Measures Used

1. Scenario Title

Scenario Title = "Rating >= " & FORMAT([Rating Threshold Value], "0") & " | Safety + " & FORMAT([Safety Improvement Value], "0%") & " | Readmission + " & FORMAT([Readmission Improvement Value], "0%")

2. High-Performing Hospitals

```
High-Performing Hospitals = VAR SelectedThreshold = SELECTEDVALUE('Rating  
Threshold'[Rating Threshold], 3) RETURN CALCULATE(COUNTROWS(Hospital),  
FILTER(Hospital, Hospital[OverallRating] >= SelectedThreshold))
```

3. Projected Safety Better

```
Projected Safety Better = [Safety Better] * (1 + [Safety Improvement Value])
```

4. Projected Readmit Better

```
Projected Readmit Better = [Readm Better] * (1 + [Readmission Improvement Value])
```

5. Best Performing Domain

```
Best Performing Domain = VAR Mort = [Mort Better %] VAR Safety = [Safety Better %]  
VAR Readm = [Readm Better %] VAR Highest = MAX(MAX(Mort, Safety), Readm) RETURN  
SWITCH(Highest, Mort, "Mortality", Safety, "Safety", Readm, "Readmission")
```

Default Scenario Results

Parameter	Value
Rating Threshold	>= 3
High-Performing Hospitals	2,000
Projected Safety Better	2,810
Projected Readmit Better	1,330
Best Performing Domain	Safety

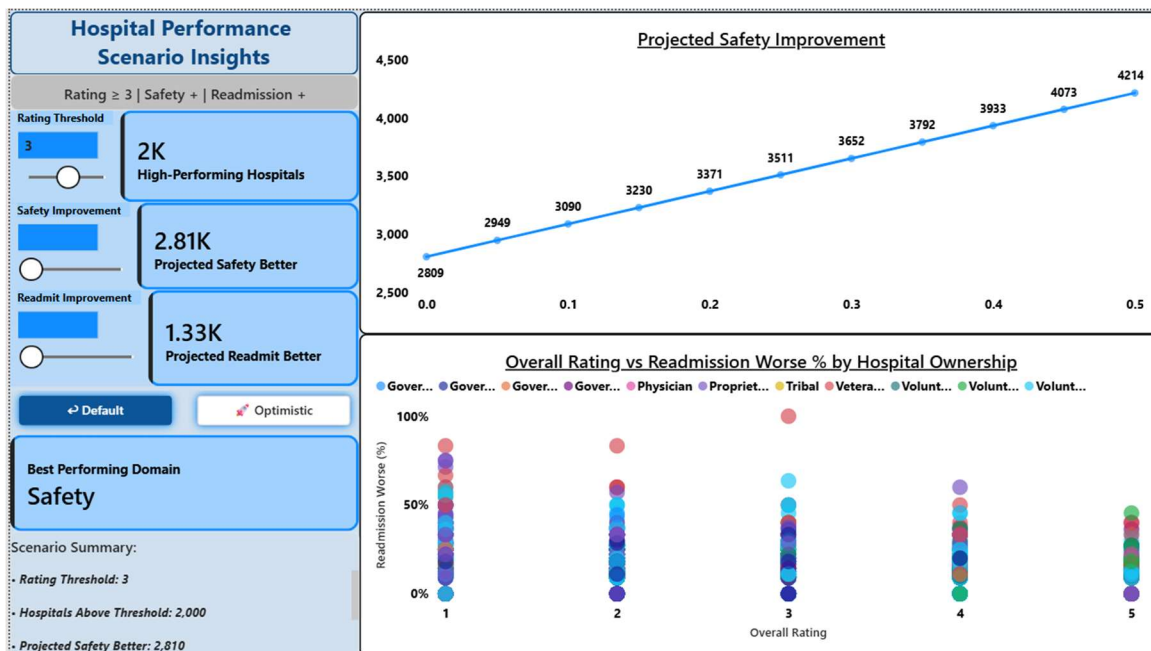
Projected Safety Improvement Curve

Improvement %	Projected Safety Better
0% (Baseline)	2,809
10%	2,949 - 3,090
20%	3,230 - 3,371
30%	3,511 - 3,652
40%	3,792 - 3,933
50%	4,073 - 4,214

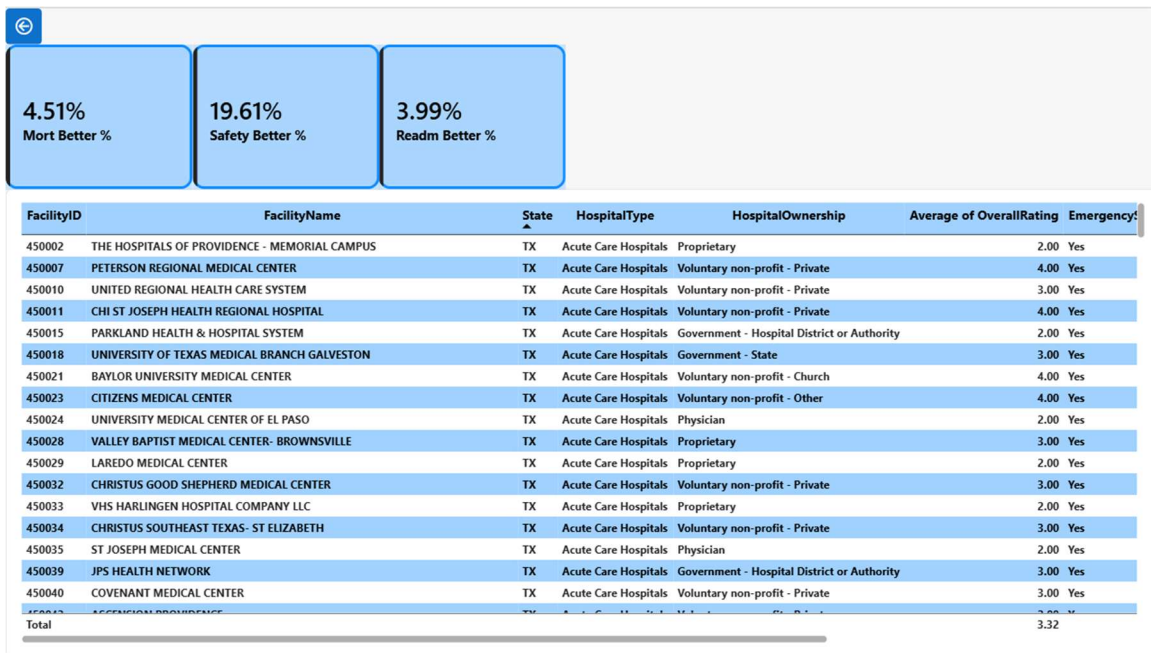
Features

- Rating Threshold slider
- Safety Improvement slider
- Readmission Improvement slider
- Best Performing Domain card
- Current vs Projected Analysis line chart
- Overall Rating vs Readmission Worse % scatter chart (by Ownership)
- Scenario Summary
- Default and Optimistic Bookmarks

Business Insight: Decision-makers can evaluate the impact of improving hospital quality metrics before implementing operational changes. A 50% safety improvement increases projected "Better" scores from 2,809 to 4,214.



Page 5: State and Hospital Detail



Key Metrics (Example: Texas)

Metric	Value
Mort Better %	4.51%
Safety Better %	19.61%
Readm Better %	3.99%
Average Overall Rating	3.32

Features

- State-level KPI cards
- Detailed hospital table with FacilityID, FacilityName, State, HospitalType, HospitalOwnership, Average of OverallRating, EmergencyServices, BirthingFriendly
- Drill-through navigation from other pages

Interactive Features

Dashboard Interactivity

The report includes several interactive Power BI capabilities.

Slicers:

- State
- Hospital Type
- Hospital Ownership
- Emergency Services
- Overall Rating
- Rating Threshold

Drill-through: Allows users to navigate from summary visuals to detailed hospital information.

Tooltips: Displays detailed hospital information when hovering over map bubbles.

Bookmarks:

- Default Scenario
- Optimistic Scenario

What-If Parameters:

- Rating Threshold
- Safety Improvement
- Readmission Improvement

Challenges Faced

Challenges

During development several challenges were encountered:

20. Missing Hospital Name field
21. Missing Patient Experience measures
22. Dynamic Scenario calculations
23. Tooltip configuration

- 24. Bookmark synchronization
- 25. Matrix formatting
- 26. Parameter integration
- 27. DAX optimization

Solutions

These issues were resolved using Power Query transformations, custom DAX measures, bookmarks, and Power BI interactive features.

Business Insights

Key Insights

- 28. Approximately 36% of hospitals have an Overall Rating of 4 or above.
- 29. Average Overall Rating across all hospitals is 3.1.
- 30. Around 41% of hospitals are Birthing Friendly.
- 31. Safety is the strongest-performing quality domain (19.61% Better).
- 32. Mortality is the weakest domain with 61.33% of measures in the "Worse" category.
- 33. Texas has the highest number of hospitals in the dataset (459).
- 34. Voluntary Non-Profit Private hospitals represent the largest ownership category (2,913).
- 35. Utah leads all states with the highest average rating of 4.3.
- 36. What-If analysis shows that improving Safety and Readmission metrics significantly increases projected high-performing hospitals.

Future Enhancements

Potential improvements include:

- 37. Real-time SQL database integration
- 38. Azure Data Factory automation
- 39. Power BI Incremental Refresh
- 40. AI-based forecasting
- 41. Predictive analytics using Python
- 42. Row-Level Security (RLS)
- 43. Mobile dashboard optimization
- 44. Power BI Apps deployment
- 45. Integration with Microsoft Fabric

Conclusion

This project successfully demonstrates how Microsoft Power BI can transform raw hospital performance data into meaningful business insights through interactive dashboards.

The solution enables healthcare administrators to monitor key performance indicators, evaluate hospital quality, compare geographic performance, and perform scenario-based forecasting using What-If Parameters. The implementation of drill-through, bookmarks, custom tooltips, and dynamic DAX measures enhances usability and supports data-driven decision-making.

Overall, the dashboard provides a scalable and professional business intelligence solution that can be extended to support real-time healthcare reporting and advanced analytics.